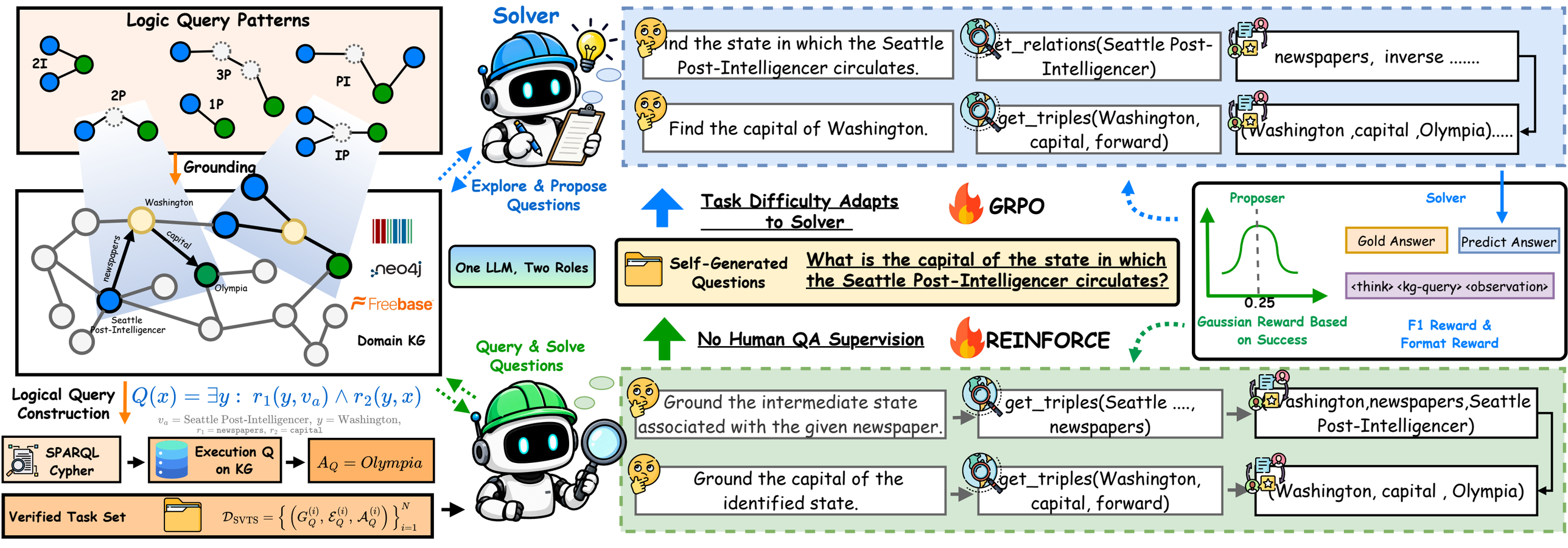

Logic Query Patterns
2I
3P
PI
2P
1P
IP
Grounding
Washington
capital
newspapers
Olympia
Seattle Post-Intelligencer
neo4j
Freebase
Domain KG
Logical Query Construction
$Q(x) = \exists y : r_1(y, v_a) \wedge r_2(y, x)$
v_a = Seattle Post-Intelligencer, y = Washington, r_1 = newspapers, r_2 = capital
SPARQL Cypher
Execution Q on KG
$A_Q = Olympia$
Verified Task Set
$\mathcal{D}_{\text{SVTS}} = \left\{ \left(G_Q^{(i)}, \mathcal{E}_Q^{(i)}, \mathcal{A}_Q^{(i)} \right) \right\}_{i=1}^{N}$
Solver
Explore & Propose Questions
One LLM, Two Roles
Query & Solve Questions
Find the state in which the Seattle Post-Intelligencer circulates.
get_relations(Seattle Post-Intelligencer)
newspapers, inverse .....
Find the capital of Washington.
get_triples(Washington, capital, forward)
(Washington ,capital ,Olympia).....
Task Difficulty Adapts to Solver
GRPO
Self-Generated Questions
What is the capital of the state in which the Seattle Post-Intelligencer circulates?
No Human QA Supervision
REINFORCE
Proposer
Solver
Gold Answer
Predict Answer
<think> <kg-query> <observation>
0.25
Gaussian Reward Based on Success
F1 Reward & Format Reward
Ground the intermediate state associated with the given newspaper.
get_triples(Seattle ...., newspapers)
ashington,newspapers,Seattle Post-Intelligencer)
Ground the capital of the identified state.
get_triples(Washington, capital, forward)
(Washington, capital , Olympia)